# Amirreza Sattarzadeh Khanehbargh

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amirreza-stz.github.io · Google Scholar · GitHub · LinkedIn

**Research interests:** Natural language processing (NLP) and machine learning (ML), centred on language models (LMs): tokenization and input representations, efficient adaptation and fine-tuning, and interpretability of what models encode. **Bringing NLP and ML to other fields:** biology (protein language models and BioNLP), the social sciences (hate speech in multilingual text), finance (market-derived sentiment), and low-resource, less-studied languages and their linguistic structure (e.g. Universal Dependencies).

## Research Experience

**Graduate Research Assistant, LifeLU (ERC Consolidator Grant)** Oct 2024 – present
*TabiLab, Boğaziçi University · PI: Prof. Arzucan Özgür*

- Study how the choice of tokenization (amino-acid, subword and evolution-aware units) interacts with fine-tuning and parameter-efficient adaptation in protein language models (PLMs), including ESM-2 models pretrained from scratch with a teammate.
- Built a benchmark of **6,500+ training runs** on **14 protein function, localization and stability tasks**. Evolution-aware units outperformed frequency-based subwords in several settings and matched or surpassed amino-acid tokens at some vocabulary sizes.
- Tested adding predicted protein-dynamics features to PLM representations. Next: interpreting the learned units, other knowledge-fusion methods and their interaction with tokenization, dynamic and task-specific tokenization.
- Regular presenter in the team's weekly reading club.

**Developer and Researcher, PARI project** Oct 2026 – present
*Hrant Dink Foundation with Boğaziçi and Sabancı universities, Istanbul*
*Funded by the EU and the Friedrich Naumann Foundation*

- Recently joined as the project's student researcher, working with its faculty, postdoctoral and PhD researchers on an open-source NLP tool for detecting hate speech and discriminatory content in Turkish and Arabic.

**Team member, ELSA Health Privacy Challenge, CAMDA 2026 at ISMB 2026** Jul 2026
*Organized by EMBL, CISPA Helmholtz Center and the University of Helsinki within the EU ELSA project*

- Equal-contribution first author with Buse Giledereli and Başar Temiz. Benchmarked single-cell RNA-seq generators, including an LLM-based one (Cell2Sentence), for biological utility against membership-inference privacy, and showed that how non-highly-variable genes are reconstructed is the main privacy lever.
- Our work was selected for a talk and presented as a talk and a poster at ISMB 2026, Washington, DC.

**Research Assistant** Jul 2023 – Sep 2024
*Iran University of Science and Technology (IUST), Tehran · Supervisor: Dr. Reza Entezari-Maleki*

- *Financial market trend analysis using FinBERT and sentiment analysis in online text data* (Feb – Sep 2024): labelled posts by the market move that follows them instead of human sentiment labels and fine-tuned domain language models on these labels; extended into the arXiv preprint.
- *Deep learning and neural networks for algorithmic trading* (Jul 2023 – Feb 2024).

## Publications and Posters

- **Sattarzadeh Khanehbargh, A.**, Suyunu, B., Dolu, Ö., and Özgür, A. (2026). *Tokenization-Aware Protein Language Modeling with Evolution-Guided Units and Biological Knowledge Fusion.* 25th European Conference on Computational Biology (**ECCB 2026**), Geneva. Poster C-P.51, presented in person (poster page).
- **Sattarzadeh Khanehbargh, A.** and Özgür, A. (2026). *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models.* Lectures on Computational Linguistics (**LCS 2026**), Bolzano, Italy. Accepted poster.
- Giledereli, B.*, Temiz, B.*, and **Sattarzadeh Khanehbargh, A.*** (2026). *Benchmarking Recent Single-Cell Synthetic Data Generation Methods on the OneK1K Dataset.* ELSA Health Privacy Challenge, **CAMDA 2026 at ISMB 2026**, Washington, DC (selected talk and poster). Preprint, arXiv:2610.31454. *Equal contribution.
- Moradi-Kamali, H., Rajabi-Ghozlou, M.-H., Ghazavi, M., Soltani, A., **Sattarzadeh, A.**, and Entezari-Maleki, R. (2025). *Market-Derived Financial Sentiment Analysis: Context-Aware Language Models for Crypto Forecasting.* Preprint, arXiv:2502.14897. 12 citations on Google Scholar, including its earlier version.

## Education

**M.Sc. in Computer Engineering** Oct 2024 – expected Jul 2027
*Boğaziçi University, Istanbul, Türkiye (English-taught)*

- Supervisor: Prof. Arzucan Özgür, Text Analytics and Bioinformatics Lab (TabiLab) · GPA 3.25 / 4.0
- Thesis (in progress): *From Protein Sequences to Biological Language Units: Frequency- and Evolution-Aware Tokenization for Protein Language Models*

**B.Sc. in Computer Engineering** Sep 2018 – Sep 2024
*Iran University of Science and Technology (IUST), Tehran, Iran*

- Final project: *Financial Market Trend Analysis using Sentiment Polarity in Online Text Data* (Dr. Reza Entezari-Maleki), extended into the arXiv preprint.

**High school diploma, Mathematics and Physics** 2014 – 2018
*SAMPAD (NODET: National Organization for Development of Exceptional Talents), selective gifted school, Bonab, Iran · ranked 1st among the school's Mathematics and Physics students*

## Industry Experience

**Python Developer (Backend and Data), Roshan-AI** Jun 2022 – Nov 2023
*Tehran · deep-learning company · joined through a competitive backend internship, then part-time developer*

- *Harf* (Persian speech recognition): built the data pipeline for crawling, cleaning and preprocessing speech–text data, growing the training data in size and quality, which improved the speech models; Django APIs for deployment.
- *Replai* (NLP comment and reply classification): Django backend.

**Junior Software Engineer (freelance)** Sep 2019 – Feb 2022
*Bonab, Iran*

- Telegram bot that crawls Iranian stock-market data and screens growth stocks; Django backend for a charity website in a team of five.

## Teaching and Mentoring

- **Teaching Assistant, IUST** (Sep 2023 – Jul 2024), seven course sections: Head TA for Data Science; Software Engineering Mentor and Scrum Master for Systems Design and Analysis; Scrum Master for Software Engineering II; TA for Advanced Programming (two sections), Fundamentals of Computer Programming, Computer Workshop.
- **Python Instructor**, independent, online (Jan 2023 – Jun 2024): introductory Python for beginners.
- **Volunteer Instructor**, Pooyandegan-e-Daneshpazhooh-e-Mehr (BonabNGO) (Summer 2023): designed and ran a free online Python and computer-engineering fundamentals course for students in the region.

## Honors, Tests and Service

- Iran's national graduate entrance exam (2024): ranked 764th in Computer Engineering and 398th in Information Technology nationwide; chose Boğaziçi instead. National university entrance exam (2018): ranked 233rd of about 160,000 (Mathematics). National Mathematical Olympiad, 2nd stage (2016).
- GRE Quantitative Reasoning: 165 / 170 (2024).
- Board member (since 2021), Pooyandegan-e-Daneshpazhooh-e-Mehr (BonabNGO), an educational NGO. Core team, Iran Innovate (Iran branch of Innovate Decentral), 2023 – 2024. Member of ISCB and AILC.

## Tools and Languages

**Tools** Python, PyTorch, Hugging Face, Weights & Biases, Linux and remote GPU servers, Git, Django
**Languages** Persian, Azerbaijani and Turkish (native or bilingual); English (full professional; English-taught M.Sc.); Arabic (elementary)

## Referees

Further references available on request.

- Prof. Arzucan Özgür, Boğaziçi University (Dept. Chair); M.Sc. supervisor, LifeLU PI · arzucan.ozgur@bogazici.edu.tr
- Dr. Reza Entezari-Maleki, IUST; B.Sc. project and research supervisor, coauthor · entezari@iust.ac.ir
- Dr. Mehrdad Ashtiani, IUST; supervised my mentoring, Scrum Master and TA roles · m_ashtiani@iust.ac.ir
- Dr. Hassan Naderi, IUST; supervised my Head TA role in Data Science · naderi@iust.ac.ir
- Dr. Maryam Khodadadi, University College London (UCL); my CAD course instructor at IUST · m.khodadadi@ucl.ac.uk
- Dr. Reza Yousefi Maragheh, Walmart Global Tech; mentor, fellow BonabNGO board member · ryousefimaragheh@acm.org
- Dr. Mohsen Mohammadi, Roshan-AI; my direct manager (Harf speech recognition) · mohammadi.mohsen@gmail.com